MAJOR PROJECT – AIML-B02

Department of Artificial Intelligence and Machine Learning

AI-Driven Enterprise Knowledge Digital Twin using Agentic AI and Hybrid RAG

Intelligent knowledge retrieval, role-based access control, and department-specific enterprise search assistants.

PROJECT GUIDE

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TEAM MEMBERS

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Introduction & Background

Multi-Document Silos

Enterprise organizations store vital intellectual property across disconnected repositories, including technical PDFs, Word policies, financial spreadsheets, and internal wikis.

Productivity Bottlenecks

Employees spend substantial working hours manually searching through unindexed departmental directories just to locate standard operational answers.

Confidentiality Boundaries

Distinct business units like Human Resources, Engineering, and Finance maintain strictly separate records requiring uncompromised data segregation.

Project Scope: Developing an intelligent enterprise digital twin assistant that interprets natural language queries, executes hybrid keyword-semantic retrieval, and enforces granular RBAC.

Problem Statement

Fragmented Repositories

Critical knowledge is scattered across hundreds of unstructured documents with zero centralized indexing or cross-reference capabilities.

Inefficient Manual Search

Standard keyword lookups fail to capture contextual semantics, forcing staff to read through entire documents manually.

Security Vulnerabilities

Lack of automated role verification risks unauthorized access to sensitive financial sheets and confidential HR employee records.

Accuracy Deficits

Generic LLMs lack enterprise grounding, leading to hallucinations and unverified citations on critical operational policies.

Core Need: A unified, secure, and context-aware knowledge retrieval platform engineered specifically for multi-department enterprise workflows.

Project Objectives & Goals

Intelligent Enterprise Assistant

Deploy a digital twin assistant capable of answering complex internal queries instantly with verifiable source citations.

Natural Language Processing

Enable conversational employee queries without requiring complex SQL syntax or rigid keyword matching strings.

Advanced Hybrid RAG Pipeline

Merge BM25 keyword precision with Qdrant vector semantic search, optimized via Reciprocal Rank Fusion and rerankers.

Role-Based Access Enforcement

Integrate strict RBAC protocols ensuring users only access documents authorized for their specific department.

Target Outcome: Maximize retrieval relevancy, minimize hallucinations, and streamline cross-departmental knowledge sharing securely.

Technology Stack



Frontend

React, Vite



Backend

Python, FastAPI



Databases

PostgreSQL, Qdrant



AI & Retrieval

GPT-4o-mini, BGE, BM25s, RRF, Cross-Encoder



Monitoring

RAGAS, LangSmith

Modern, scalable technology stack engineered for high-performance enterprise RAG.

System Architecture Flow

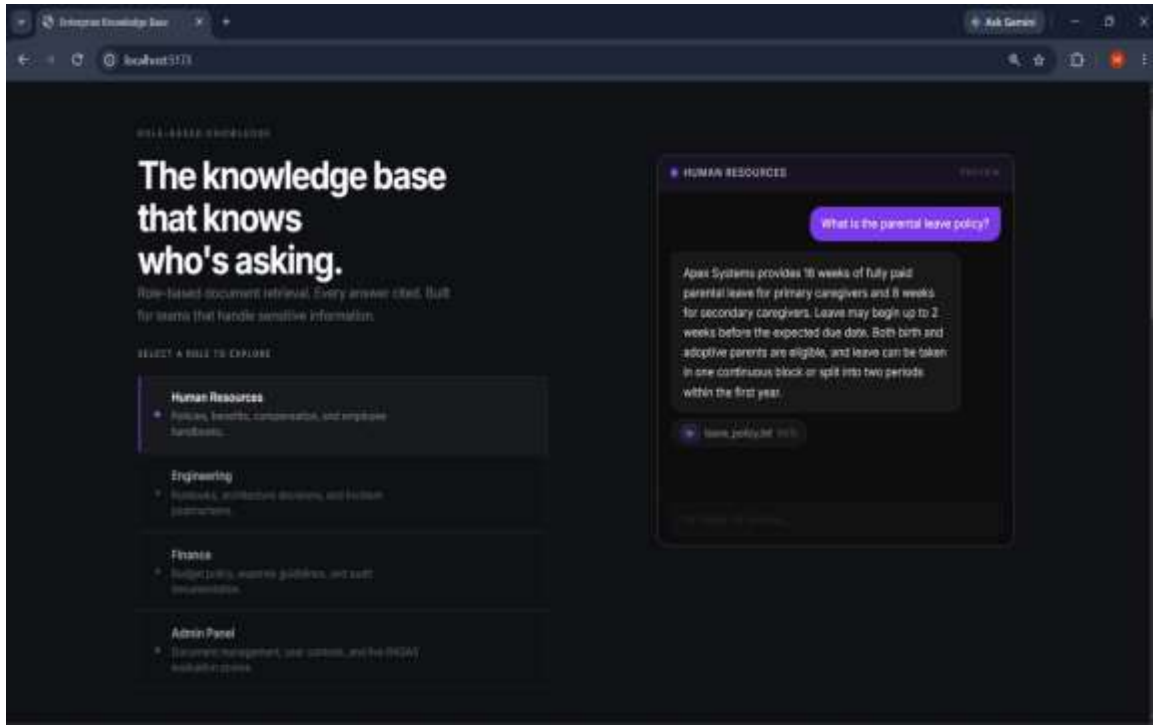


PostgreSQL: User accounts & document metadata

Qdrant: High-dimensional vector embeddings

Knowledge Source: Enterprise department documents

Application Home Page



Project Introduction
Centralized portal introducing the enterprise digital twin assistant.

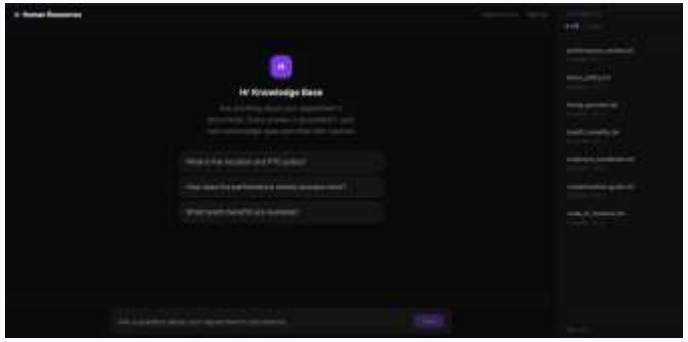
Department Selection
Dedicated navigation cards for HR, Engineering, and Finance workspaces.

Admin Panel Access
Secure authentication gateway leading to administrative controls.

Technology Overview
Summary of underlying AI models, vector databases, and RAG frameworks.

Localhost landing page providing intuitive user onboarding and workspace routing.

Department-Wise Chat Interfaces



HR Workspace

Specialized interface for employee handbooks, leave policies, onboarding procedures, health benefits, and performance review guidelines.



Engineering Workspace

Technical repository assistant for system architectures, deployment playbooks, API documentation, and security protocols.



Finance Workspace

Secure financial assistant providing instant insights into departmental budgets, expense reports, and compliance guidelines.

Dept Authentication

Natural Language Chat

Prompt Suggestions

Source Document Viewer

Role-Based Access Control (RBAC) Architecture

Departmental Data Isolation

Ensures HR personnel access personnel records only. Engineering and Finance staff are strictly partitioned from querying sensitive cross-department confidential filings.

Secured Vector Filtering

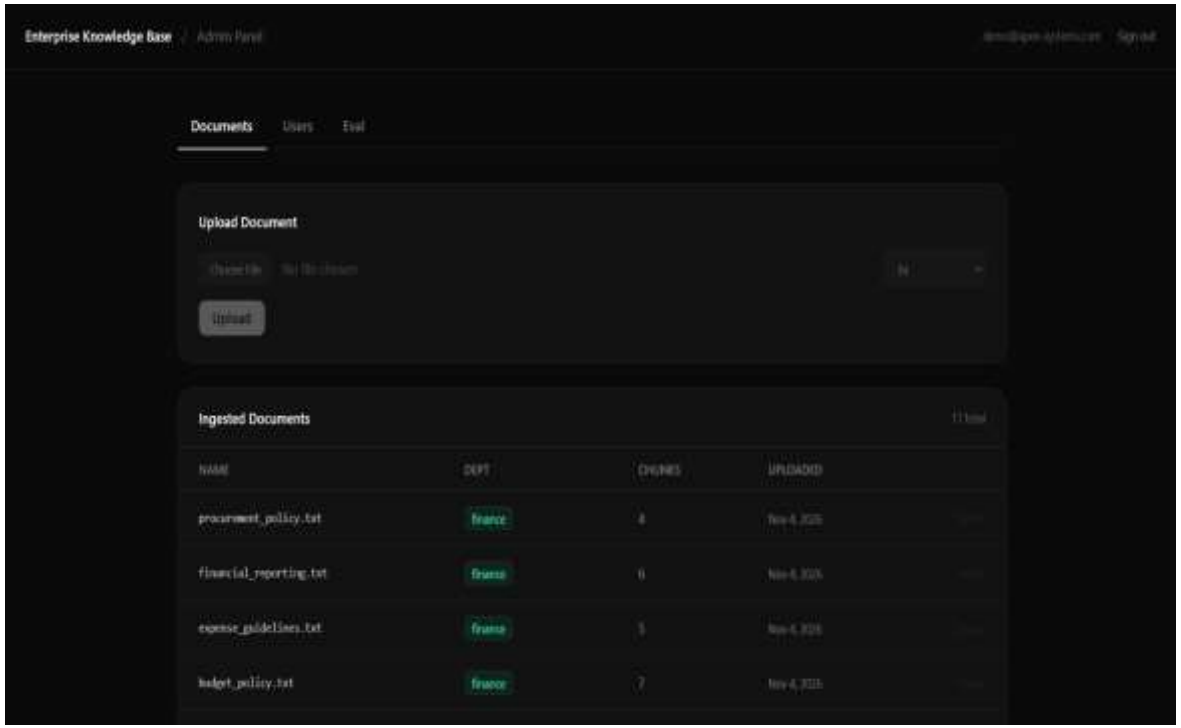
The backend search pipeline injects mandatory metadata filters into Qdrant vector queries, restricting retrieval exclusively to documents matching the user's role.

Administrator Governance

System administrators retain full auditing oversight, managing user account activations, role promotions, and global permission parameters.

Enterprise Guarantee: Zero data leakage between departments with verifiable audit trails for all queries.

Admin Panel Control



Document Management

Inspect uploaded enterprise files, department tagging, chunk counts, and file uploads/deletions.

User Account Control

Manage employee accounts, assign departmental roles, and toggle user active/inactive statuses.

RAG Evaluation Suite

Monitor Faithfulness, Answer Relevancy, Context Recall, and Context Precision metrics via RAGAS.

Centralized administrative dashboard ensuring high retrieval fidelity and secure access.

Hybrid RAG Workflow

STEP 01

Ingestion

Upload, extraction, chunking, & vector embedding.

STEP 02

Retrieval

BM25 keyword search + Qdrant semantic vector search.

STEP 03

Fusion & Rerank

RRF result merging & Cross-Encoder precision reranking.

STEP 04

Generation

GPT-4o-mini produces natural language answers with sources.

BM25: Keyword Precision

Vector Search: Semantic Depth

RRF: Robust Fusion

Reranker: High Relevancy

Current Status & Future Work

Currently Explored & Built

- Local development environment fully configured
- Landing page and workspace routing operational
- HR, Engineering, and Finance UI panels active
- Admin panel interface built with document lists

Under Active Testing

- End-to-end hybrid RAG question-answering pipeline
- Source citation verification and preview rendering
- Role-based access control permission validation
- RAGAS evaluation metrics benchmarking

Future Expansion

- Complete FastAPI backend and vector db integration
- Fine-tune Cross-Encoder reranker accuracy
- Implement real-time document chunk previews
- Deploy containerized production build (Docker)

Solid architectural foundation established with active optimization of the RAG pipeline.

Conclusion

The project successfully establishes a centralized AI-powered enterprise knowledge assistant utilizing Hybrid RAG.

Role-based access control guarantees strict departmental information security and compliant data sharing.

Thank You! — AIML-B02 (Guide: Dr. K. Gnanendra)